

Spectroscopy as a Probe of Spacetime Foam

A. G. Kim*

Physics Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

P. E. Nugent

*Applied Mathematics and Computational Research Division,
Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA*

L. Wang

Department of Physics and Astronomy, 4242 TAMU, College Station, Texas 77843-4242, USA

(Dated: May 22, 2026)

Quantum gravity predicts that spacetime fluctuates stochastically at the Planck scale — a “foam” that imprints a tiny achromatic arrival-time jitter on light from distant sources.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the proper distance to the source, α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Spacetime foam accumulates a cumulative phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

We develop a framework based on the first-order coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. The first-order coherence – the field–field autocorrelation, equivalently the fringe visibility at delay τ – inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows, extending the binary Lieu–Hillman erasure criterion [7] ($\sigma_\phi \gtrsim 1$) to a continuous curve and recovering the Strehl and fringe-visibility tests as limiting cases of the same statistic.

The remainder of the paper is organized as follows.

* agkim@lbl.gov

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for virtually all known astrophysical sources) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (3)$$

In the geometric-optics limit the path-length shift $\Delta \ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity is a key discriminant between foam and conventional propagation effects.

B. Spatial and Temporal Fluctuations of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — quantum foam models posit that each Planck-scale cell along the line of sight fluctuates in time. Two photons traversing the same cell at coordinate times separated by τ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\text{foam}}$ but decorrelated for $\tau \gg \tau_{\text{foam}}$, where τ_{foam} is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_P$ cells along the line of sight. The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. If τ_{foam} is of order the Planck time, $\ell_P/c \approx 5 \times 10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\text{foam}} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all τ of interest and every C_α -dependent temporal observable vanishes: the finite-lag Siegert discrepancy $\mathcal{S}(\tau)$ is suppressed by $[1 - C_\alpha(\tau)] \rightarrow 0$ (Eq. (??)), and the single-ray $g_{\text{obs}}^{(1)}$ recovers its intrinsic value because the foam suppression factor $\mathcal{F}_{\text{foam}} = e^{-\sigma_\phi^2[1-C_\alpha]} \rightarrow 1$ (Eq. (24)). In this quasi-static limit the HBT framework loses sensitivity altogether and the foam manifests only through the imaging-channel $g^{(1)}$ suppression on time-averaged exposures. The full spatiotemporal covariance between two rays separated by transverse distance b takes the factorized form

$$\langle \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}) \Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}') \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \quad (4)$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (so $b = 0$):

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (5)$$

This product form — separating the total variance from the normalized spatiotemporal shape — is a structural assumption, not a consequence of Eq. (1), that defines a tractable class of foam models within which the analytic framework is exact. The parameters A_α , $C_\alpha(\tau; D)$, τ_{foam} , $r_\perp$, and Φ_{eff} parameterize distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and each governs a distinct aspect of the signal's detectability. The factorization is consistent with phenomenologies in which the longitudinal path-length variance, the temporal decorrelation, and the transverse coherence arise from statistically independent mechanisms — e.g. the simple cellular pictures of Refs. [4–6] where each Planck-scale cell contributes an independent random kick.

The factorization is *not* generic: the pixellon model of Lee, Zurek & Chen [10] has long-range angular correlations whose spatiotemporal covariance does not reduce cleanly to $\sigma_\ell^2(D) C_\alpha(\tau) \Phi(b/r_\perp)$; the framework here applies to that model only as a heuristic approximation in the regime where transverse correlations have decoupled.

Observationally, factorization predicts that the extended-source Siegert discrepancy is suppressed relative to its point-source value by the angular-size factor $\Phi_{\text{eff}}(r_{\perp})$: $\mathcal{S}_{\text{ext}}(\tau) \approx \Phi_{\text{eff}} \mathcal{S}_{\text{pt}}(\tau)$ in the weak-foam regime (Section ??), independent of both τ and D . A dependence of this ratio on lag or source redshift would falsify the factorization ansatz and indicate that a non-factorizing model (e.g. pixellon) is required. The single observable developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ — descends from Eq. (1), with Sections III–?? probing its longitudinal projection (single-ray temporal jitter) and Section ?? its transverse projection (cross-ray Φ_{eff} suppression).

C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t + \tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time t :

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}'). \quad (6)$$

For fixed D , stationarity implies the statistics depend only on τ and the transverse separation $b(\boldsymbol{\theta}, \boldsymbol{\theta}') = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}) - \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}')]^2 \rangle \\ &= \frac{1}{c^2} \left[\sigma_{\ell}^2 + \sigma_{\ell}^2 - 2\sigma_{\ell}^2 C_{\alpha}(\tau) \Phi\left(\frac{b}{r_{\perp}}\right) \right] \\ &= \frac{2\sigma_{\ell}^2(D)}{c^2} \left[1 - C_{\alpha}(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_{\perp}}\right) \right]. \end{aligned} \quad (7)$$

For $b \gg r_{\perp}$, $\Phi \rightarrow 0$ and the variance becomes $2\sigma_{\ell}^2/c^2$ at all lags, including $\tau = 0$ — a new effect absent in the point-source case, and one that is independent of the temporal fluctuation structure of the foam (i.e. independent of C_{α}). In the dynamical limit $C_{\alpha} \rightarrow 0$ this result holds for all separations b , independently of Φ .

By the central limit theorem (CLT), $\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \sim \mathcal{N}(0, \sigma_{\Delta}^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}')), \quad (8)$$

where σ_{Δ}^2 is taken from Eq. (7).

The Gaussian approximation follows from the CLT: for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_{\text{P}} \gtrsim 10^{46}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For the cross-ray case the additional condition $b \gg r_{\perp}$ is assumed, and is satisfied in the fully-resolved limit $\Phi_{\text{eff}} \rightarrow 0$ adopted in Section ??.

III. FIRST-ORDER COHERENCE

A. Point-Source Differential Jitter

In this section we restrict to a point source ($b = 0$), for which the cross-ray differential jitter of Eq. (6) reduces to the single-ray differential jitter

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t), \quad (9)$$

with variance obtained from Eq. (7) at $\Phi(0) = 1$:

$$\sigma_{\Delta}^2(\tau; D) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - C_{\alpha}(\tau; D)]. \quad (10)$$

Although this section restricts to a point source for notational compactness, this single-ray variance applies equally to extended sources at finite lag $\tau \neq 0$: under the factorized covariance Eq. (4), the temporal factor $C_{\alpha}(\tau; D)$ is identical for all ray pairs, so cross-ray transverse structure enters only at $\tau = 0$ through Φ_{eff} (Section ??).

B. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (11)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (12)$$

Since δt is independent of E_{src} , averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (13)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

C. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}, \quad (14)$$

where τ_L is the coherence time. Evaluating Eq. (13) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = e^{\sigma_{\Delta}^2/2\tau_L^2} \cdot \frac{1}{2} \left[e^{-\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L - \tau/\sigma_{\Delta}}{\sqrt{2}}\right) + e^{\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L + \tau/\sigma_{\Delta}}{\sqrt{2}}\right) \right]. \quad (15)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \tau_L$, the Gaussian average of $e^{-|\tau - \Delta|/\tau_L}$ over $\Delta \sim \mathcal{N}(0, \sigma_{\Delta}^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2\tau_L^2} + \mathcal{O}(\sigma_{\Delta}^4) \right]. \quad (16)$$

Here $\sigma_{\Delta}(\tau)$ is τ -dependent through $C_{\alpha}(\tau)$, and the expansion is in $\sigma_{\Delta}(\tau)/\tau_L$ at fixed τ . The correction is positive because $e^{-|\tau|/\tau_L}$ is convex and Jensen's inequality applies. The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved exactly.

For a Voigt profile $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width τ_L (from optically-thick electron scattering or BLR turbulent broadening) convolved with a Gaussian of width τ_G (from bulk Doppler motions) — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde{\sigma}_1^2 = \frac{\sigma_{\Delta}^2 \tau_G^2}{\sigma_{\Delta}^2 + \tau_G^2}, \quad \mu_1(\tau) = \frac{\tau_G^2 \tau}{\sigma_{\Delta}^2 + \tau_G^2}, \quad (17)$$

giving, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_{\Delta}} e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)} e^{\tilde{\sigma}_1^2/2\tau_L^2} \frac{1}{2} \left[e^{-\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) \right]. \quad (18)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \min(\tau_L, \tau_G)$, expanding Eq. (13) to second order in Δ and averaging over the Gaussian jitter gives the Voigt analogue of Eq. (16):

$$g_{\text{obs}}^{(1)}(\tau) \approx g_{\text{src}}^{(1)}(\tau) \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2} \left(\left(\frac{1}{\tau_L} + \frac{\tau}{\tau_G^2} \right)^2 - \frac{1}{\tau_G^2} \right) + \mathcal{O}(\sigma_{\Delta}^4) \right], \quad (19)$$

which reduces to Eq. (16) in the purely Lorentzian limit $\tau_G \rightarrow \infty$. Unlike the Lorentzian case, the correction factor can change sign: for $\tau < \tau_G \sqrt{1 - \tau_G/\tau_L}$ the second-order term is negative, so foam jitter *suppresses* short-lag coherence. Note that both closed forms in the weak-jitter limit share the general product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) F_{\text{foam}}(\tau)$.

D. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, $E_{\text{src}}(t) = A(t) e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c \Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t - \delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\text{obs}}(t) \approx A(t) e^{-i\omega_0 t} e^{i\omega_0 \delta t(t)}. \quad (20)$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is ≈ 0.28 , so the exact convolution Eq. (15) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Section ??), so Eq. (22) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0 ; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \quad (21)$$

This holds when $\tau \gg \tau_{\text{foam}}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\text{foam}} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (10):

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \exp(-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]), \quad (22)$$

where the rms accumulated phase follows from the Ng–Perlman path-length model (1) via the phase conversion $\sigma_\phi = (2\pi/\lambda_{\text{obs}}) \sigma_\ell(D_C)$, with D_C the line-of-sight comoving distance [4]:

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}. \quad (23)$$

We write the foam suppression factor — the quantity appearing in the product structure of Section III C — as

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \quad (24)$$

so $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \mathcal{F}_{\text{foam}}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2 [1 - C_\alpha(\infty; D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ is compatible with the weak-jitter expansion of Eq. (16) — the former controls envelope displacement, the latter the carrier-phase suppression — and when neither applies the full convolution Eq. (13) must be evaluated.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (13) with closed forms Eqs. (15)–(18) — carries direct spectroscopic content: for any wide-sense stationary (WSS) field, the normalized power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\{g_{\text{obs}}^{(1)}(\tau)\}(\nu). \quad (25)$$

The theorem applies here for two reasons. First, astronomical emission lines have finite linewidth: their spectra fall to zero at both low and high frequencies, so $\Gamma_{\text{src}}(\tau) \rightarrow 0$ and $\Gamma_{\text{src}} \in L^1$; since the foam factor is bounded, $\Gamma_{\text{obs}} \in L^1$ as well and the Fourier transform is well defined. Second, $E_{\text{obs}}(t)$ is wide-sense stationary: the foam-induced jitter $\delta t(t)$ is a stationary process (Eq. (5), Section II) that is statistically independent of the source, so $\Gamma_{\text{obs}}(\tau)$ — given in general by Eq. (13) — depends only on the lag τ , provided $T_{\text{int}} \gg \tau_{\text{foam}}$ so that the foam statistics are ergodically sampled. In the further limit where the foam decorrelates on long time scales ($C_\alpha(\tau) \rightarrow 0$ as $|\tau| \rightarrow \infty$, the dynamical regime), the foam factor acquires a nonzero asymptote $\mathcal{F}_{\text{foam}}(\tau) \rightarrow e^{-\sigma_\phi^2}$, generating a distributional $\delta(\nu)$ contribution that encodes the fraction $e^{-\sigma_\phi^2}$ of source flux propagating phase-coherently.

When the lag-domain product structure $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \mathcal{F}_{\text{foam}}(\tau)$ holds—the weak-jitter and narrow-band regimes—the convolution theorem applied to Eq. (25) yields

$$S_{\text{obs}}(\nu) = S_{\text{src}}(\nu) * \mathcal{K}_{\text{foam}}(\nu). \quad (26)$$

i.e. the intrinsic spectrum is convolved by a kernel $\mathcal{K}_{\text{foam}}(\nu) \equiv \mathcal{F}\{\mathcal{F}_{\text{foam}}\}(\nu)$ that depends on both the foam (through σ_ϕ^2 , C_α , τ_{foam}) and the source line.

B. Connection to Fringe Visibility and Strehl Ratio

The first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (24). The fringe visibility and Strehl ratio are the *spatial* counterparts: they probe phase variations between two points in the telescope aperture plane separated by a baseline B , with the suppression governed by an aperture-plane coherence function $C_\alpha^{\text{sp}}(B; D)$.

The structural equivalence between the temporal and spatial cases is not merely formal: in both, the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable, with C_α (respectively C_α^{sp}) setting how that variable's variance grows with separation. Temporally, the relevant random variable is the differential jitter $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$ between two arrival times along a single ray, with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (10)). Spatially, the analogous random variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (27)$$

Applying the Gaussian characteristic-function identity of Eq. (21) in each domain yields suppression factors of identical form—one expressed in lag, the other in baseline:

$$\mathcal{F}_{\text{foam}}^{\text{sp}}(B) \equiv \exp\{-\tfrac{1}{2}\sigma_{\Delta\phi}^2(B)\} = \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (28)$$

mirroring $\mathcal{F}_{\text{foam}}(\tau)$ of Eq. (24) under $C_\alpha(\tau; D) \leftrightarrow C_\alpha^{\text{sp}}(B; D)$. The shared subscript α further reflects that the same cumulative growth law of Eq. (1) underlies both decorrelations. As with its temporal counterpart, $\mathcal{F}_{\text{foam}}^{\text{sp}}$ runs from unity at zero baseline ($C_\alpha^{\text{sp}}(0; D) = 1$, the two points sample identically phased wavefronts) to maximum suppression $e^{-\sigma_\phi^2}$ in the resolved limit where $C_\alpha^{\text{sp}} \rightarrow 0$ (the two points sample independent foam realizations).

The two-element interferometric fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \mathcal{F}_{\text{foam}}^{\text{sp}}(B) = V_{\text{src}}(B) \exp\{-\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]\}, \quad (29)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}|$ is evaluated at angular scale λ/B . The Strehl ratio of a single aperture of diameter D_{tel} is the same statistic with B replaced by the internal aperture baseline: when $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$ the aperture-plane phases are independent draws from the single-ray distribution and the Maréchal formula gives [4]

$$S \simeq e^{-\sigma_\phi^2}. \quad (30)$$

Strehl ratio and fringe visibility therefore measure the *same* underlying phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the resolved limit, differing only in whether what part of the point-spread-function is being considered. This unification places the published imaging constraints [4, 7–9] on a common footing within the present framework.

Equation (29) also extends the binary Lieu–Hillman [7] erasure criterion to a continuous suppression curve. Their condition $\Delta\phi \gtrsim 2\pi$, with $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$, is precisely $\sigma_\phi \gtrsim 1$ —the saturation limit $\mathcal{F}_{\text{foam}}^{\text{sp}} \rightarrow e^{-\sigma_\phi^2} \rightarrow 0$ of Eq. (28). The shape of the continuous curve, observed as a function of baseline, encodes the aperture-plane coherence profile $C_\alpha^{\text{sp}}(B; D)$, in direct analogy to how $g_{\text{obs}}^{(1)}(\tau)$ encodes the temporal decorrelation $C_\alpha(\tau; D)$. Two observational handles follow from this correspondence: (i) the chromatic scaling $\sigma_\phi \propto 1/\lambda$ implies that constraints tighten rapidly with decreasing wavelength—a fact exploited by Ng & Perlman [4] (their Table 1) in moving from optical through X-ray and γ -ray observations, and mirrored in Section V A, where the same saturated-limit suppression $e^{-\sigma_\phi^2}$ that defines the Strehl ratio (Eq. (30)) is reinterpreted as the coherent fraction of a spectral line, yielding a spectroscopic counterpart of the Strehl bound at X-ray wavelengths; (ii) baseline-resolved visibility measurements of a single target can in principle probe the aperture-plane coherence scale, complementary to the temporal decorrelation scale accessed through $g_{\text{obs}}^{(1)}(\tau)$.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are blurred, in direct analogy to the blurring of the point-spread function in imaging. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- y distortions. In this section we consider these two possibilities.

A. BzK4892

The two-component line-profile decomposition of Eq. (26) admits a direct application as a spectroscopic analogue of the Strehl-ratio bound [4]: just as the observation of a diffraction-limited point-spread function at wavelength λ and distance D requires the Strehl ratio $\mathcal{S} \simeq e^{-\sigma_\phi^2}$ (Eq. (30)) to remain of order unity, i.e. $\sigma_\phi^2 \lesssim \mathcal{O}(1)$ (§3 and Eq. (4) of Ref. [4]), the observation of sharp spectral features requires the same condition: in the dynamical foam limit the lag-asymptote $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$ of Eq. (24) plays the role of a *spectral Strehl* — the fraction of source flux delivered into the coherent line core $e^{-\sigma_\phi^2} S_{\text{src}}(\nu)$ of Eq. (26) — and for $\sigma_\phi^2 \gtrsim 1$ this coherent core is suppressed below the incoherent pedestal so that crisp line profiles cease to be detectable. The two arguments are structurally identical: each is a single-number, saturated-limit test of the coherent fraction $e^{-\sigma_\phi^2}$, differing only in whether that fraction is read off the on-axis PSF intensity or off the spectral line core. They share the same threshold criterion and the same parametric dependence on $\{D, \lambda, \alpha\}$.

BzK 4892 is a massive ($M_\star \approx 5 \times 10^{10} M_\odot$), infrared-luminous, dust-obscured star-forming *BzK* galaxy in the CDF-S at spectroscopic redshift $z_s = 2.578$, hosting a heavily obscured, intrinsically quasar-luminous ($L_{2-10 \text{ keV}} \gtrsim 10^{44.5} \text{ erg s}^{-1}$) AGN behind a Compton-thick absorber with $N_{\text{H}} \sim 10^{24.5-25} \text{ cm}^{-2}$ [11]. Because the direct hard X-ray continuum is suppressed by photoelectric absorption and Compton down-scattering, the X-ray spectrum is reflection-dominated and the 6.4 keV Iron K α fluorescence line—produced by the obscurer reprocessing the buried AGN continuum—emerges with exceptionally large rest-frame equivalent width, $\text{EW} \simeq 2.3 \text{ keV}$, detected at $\gtrsim 4\sigma$ in the *Chandra* 4 Ms CDF-S exposure [11]. The redshift is independently confirmed by multiple narrow rest-UV emission lines (Ly α , C IV, He II, C III) detected in VLT/FORS spectroscopy [12, 13]. Adopting the flat Λ CDM cosmology used in Ng–Perlman ($\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the line-of-sight comoving distance is $D_C(z_s = 2.578) \approx 6.1 \text{ Gpc} \approx 1.88 \times 10^{26} \text{ m}$. The spectroscopically detected Iron K α 6.4 keV emission line, redshifted to an observed energy of $6.4/(1+z_s) \approx 1.79 \text{ keV}$ ($\lambda_{\text{obs}} \approx 6.93 \times 10^{-10} \text{ m}$), provides the probe wavelength for this estimate; the short rest-frame wavelength ($\lambda_{\text{rest}} \approx 1.94 \text{ \AA}$) makes the foam variance $\sigma_\phi^2 \propto \lambda_{\text{obs}}^{-2}$ much larger than for any optical line at the same distance, yielding the most constraining spectroscopic bound available at this redshift.

The threshold condition $\sigma_\phi^2 \lesssim 1$ for the coherent line core of Eq. (26) to remain detectable above the foam-broadened pedestal — the spectral-Strehl criterion introduced above, mirroring the imaging Strehl bound of Eq. (30) — depends on the intrinsic source profile $S_{\text{src}}(\nu)$ only through its characteristic width $\Delta\nu_{\text{src}}$, and not through its detailed functional form. For the fluorescent Iron K α line in BzK 4892 the natural width and fine-structure splitting are negligible compared to the velocity broadening of the Compton-thick reprocessor ($\text{FWHM} \sim 10^3\text{--}10^4 \text{ km s}^{-1}$, i.e. $\Delta\nu_{\text{src}}/\nu_{\text{src}} \sim 10^{-3}\text{--}10^{-2}$), and the line is in any case instrumentally unresolved at *Chandra*/ACIS spectral resolution. We therefore adopt the narrow-line idealisation $\Delta\nu_{\text{src}} \ll \nu_{\text{src}}$, under which the coherent core and the foam-broadened pedestal of Eq. (26) are spectrally separable; profile refinements (Voigt wings, Compton shoulder) would shift the threshold $\sigma_\phi^2 \lesssim 1$ by at most factors of order unity.

In the dynamical foam limit the suppression factor Eq. (24) reduces to $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$, which is the coherent fraction of the observed line profile via Eq. (26). Detecting the Iron K α feature requires $e^{-\sigma_\phi^2} \gtrsim e^{-1}$, i.e. $\sigma_\phi^2 \lesssim 1$. At this threshold the coherent line core carries fraction $e^{-1} \approx 37\%$ of the intrinsic flux — the minimum for a line detection not dominated by the broad pedestal, and intentionally conservative relative to the Maréchal criterion $\sigma_\phi^2 \lesssim 0.2$. The cumulative phase variance follows from the Ng–Perlman path-length model (1) via the global phase conversion $\sigma_\phi = 2\pi\sigma_\ell(D_C)/\lambda_{\text{obs}}$. The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ underlying Eq. (26) is satisfied here: at $\alpha = 0.59$ one has $\sigma_\ell \approx 1.2 \times 10^{-10} \text{ m}$ while the X-ray coherence length $c\tau_c = \lambda_{\text{obs}}^2/(2\pi\Delta\lambda) \approx 3 \times 10^{-8} \text{ m}$ for $\text{FWHM} \sim 10^3 \text{ km s}^{-1}$, giving $\sigma_\ell/(c\tau_c) \approx 4 \times 10^{-3}$; the envelope approximation is accurate to better than 10^{-4} . The phase variance, reproducing Ng & Perlman [4] Eq. (4), is

$$\sigma_\phi^2 \simeq (2\pi)^2 \frac{D_C^{2(1-\alpha)} \ell_P^{2\alpha}}{\lambda_{\text{obs}}^2}, \quad (31)$$

i.e., Eq. (23) with $A_\alpha = 1$ as in Ref. [4], where D_C is the comoving distance per Ng & Perlman Eq. (5). Imposing the threshold $\sigma_\phi^2 \lesssim 1$ and inverting,

$$\alpha \gtrsim \frac{\ln(2\pi D_C/\lambda_{\text{obs}})}{\ln(D_C/\ell_P)}, \quad (32)$$

the spectroscopic analogue of Eq. (8) of Ref. [4]. Inserting the numerical values $D_C \approx 1.88 \times 10^{26} \text{ m}$, $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10} \text{ m}$, and $\ell_P = 1.6 \times 10^{-35} \text{ m}$ gives $\ln(2\pi D_C/\lambda_{\text{obs}}) \approx 83.4$ and $\ln(D_C/\ell_P) \approx 140.6$, hence $\alpha \gtrsim 0.59$.

This bound, $\alpha \gtrsim 0.59$, is substantially tighter than the optical entry of Ref. [4] Table 1. Applying the present threshold $\sigma_\phi^2 \lesssim 1$ (rms phase $\lesssim 1 \text{ rad}$) at $\lambda = 5000 \text{ \AA}$ and $D = 10 \text{ Gpc}$ gives $\alpha \gtrsim 0.55$, compared to their $\alpha > 0.53$

obtained with the equivalent threshold $\sigma_\ell \lesssim \lambda/(2\pi)$; both reflect the same physics but differ by the 2π convention in the phase definition. The improvement from 0.55 to 0.59 is not incidental: because $\sigma_\phi^2 \propto \lambda_{\text{obs}}^{-2}$, using the X-ray FeK α line at $\lambda_{\text{obs}} \approx 6.93 \times 10^{-10}$ m instead of a typical optical line at $\sim 5 \times 10^{-7}$ m increases the variance by a factor $\sim 5 \times 10^5$, making the threshold $\sigma_\phi^2 \lesssim 1$ far more constraining. Within the X-ray band, the bound is insensitive to the exact line chosen: substituting any X-ray line at the same redshift shifts α by $|\Delta\alpha| \lesssim 0.01$ ($\partial\alpha/\partial \ln \lambda_{\text{obs}} = -1/\ln(D_C/\ell_P) \approx -7 \times 10^{-3}$, so a factor-of-two energy shift changes α by only 0.005). The result does not rule out the holographic value $\alpha = 2/3$, consistent with all current optical and infrared imaging constraints, including those of Ref. [9] and the GRAVITY interferometric observation of 3C 273 [?].

Methodologically the test is independent of imaging: it operates entirely in the spectral domain and probes line-profile contrast rather than on-axis PSF intensity, providing an orthogonal cross-check on Strehl-based constraints that does not require a diffraction-limited optical system or a resolved reference PSF.

Three caveats temper this estimate. First, the argument applies only in the dynamical foam regime in which the core-plus-pedestal decomposition of Eq. (26) holds, and in the narrow-band regime $\sigma_\ell/(c\tau_c) \ll 1$ required for the product structure of Eq. (26) to hold; both conditions are verified for the BzK 4892 parameters (see above). In the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$ the spectroscopic signature vanishes and no constraint can be placed.

This dynamical-versus-quasi-static dichotomy mirrors the exposure-time dependence of the imaging Strehl: a single short exposure ($T_{\text{exp}} \ll \tau_{\text{foam}}$) sees a frozen common-mode phase and recovers $\mathcal{S} \approx 1$, whereas a long-exposure or ensemble-averaged Strehl ($T_{\text{exp}} \gg \tau_{\text{foam}}$) recovers the full suppression $e^{-\sigma_\phi^2}$. A spectroscopic observation, integrating over astronomically long times, is automatically in the latter regime whenever τ_{foam} is finite. Second, the threshold $\sigma_\phi^2 \lesssim 1$ is intentionally conservative; a likelihood-based fit of the foam pedestal fraction $(1 - e^{-\sigma_\phi^2})$ to high-resolution spectra of BzK 4892 or similar high- z targets would refine the bound but lies beyond the scope of this estimate. Third, Eq. (31) uses the Ng–Perlman cumulative scaling with comoving distance D_C directly (Eq. (23)), and the bound quoted here is directly comparable to Ref. [4] Table 1.

VI. CMB

VII. CONCLUSIONS

A single second-order coherence observable of spacetime foam emerges from this framework: the finite-lag Siegert discrepancy $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$, present when foam fluctuates on timescales shorter than the optical coherence time ($\tau_{\text{foam}} \lesssim \tau_L$). No foam-induced zero-lag bunching suppression arises in this model: at $\tau = 0$ the same-direction jitter increments vanish identically and $g_{\text{obs}}^{(2)}(0) = 2$ for any extended source viewed by a single-mode detector (Section ??; Appendix A). Its value lies not in raw sensitivity — the signal is at $\sim 10^{-16}$ for the holographic $\alpha = 2/3$ model, well below current reach — but in three structural advantages over existing fringe-visibility and Strehl-ratio tests: (i) $\mathcal{S}(\tau)$ is constructed directly from the data, with $g_{\text{src}}^{(1)}(\tau)$ obtained from the source spectrum via Wiener–Khinchin, so a detection requires no separate source model; (ii) the angular-size suppression factor $\Phi_{\text{eff}}(r_\perp) \in [0, 1]$ (Section ??) makes extended targets weaker than compact ones but converts source-size variability across the target ensemble into a probe of the foam transverse coherence length $r_\perp$, a parameter inaccessible to single-target measurements; (iii) the framework remains sensitive to foam models with long-range transverse correlations that evade imaging tests entirely [10], and supplies a chromatic discriminant against plasma scattering through the $\sigma_\phi \propto 1/\lambda$ scaling.

The framework identifies high-priority targets: intrinsically compact emitters that saturate the upper bound regardless of $r_\perp$ (narrow-line Seyfert 1 cores, lensed-quasar BLRs), and high-flux transients (Type II n supernovae, LBV giant eruptions) whose photon-rate advantage compensates for the modest cosmological lever arm at kpc–Mpc distances. A standing rapid-response HBT plan — covering η Car, any galactic core-collapse supernova, and any nearby ($D \lesssim 30$ Mpc) Type II n identified by transient surveys — would let the framework exploit such high-yield events without prior allocation; transient targets with substantial angular extent contribute to the $r_\perp$ discriminant via Φ_{eff} . Strongly-lensed quasars (Section ??; J1330–0905, comparable to the NLS1 baseline after cosmological correction) round out the persistent target class.

ACKNOWLEDGMENTS

This work was supported in part by the Director, Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.

The authors used Claude (Anthropic) and Valency Systems to assist with drafting and editing portions of this manuscript along with detailed reference searches. All content was reviewed and verified by the authors, who take full responsibility for the work.

We thank Robin Kaiser for helpful comments.

Appendix A: Pedestrian derivation of $g_{\text{obs}}^{(2)}(\tau)$

This appendix carries out the four-point function calculation underlying Section ?? step by step. The aim is to expose how the random variables (X, Y) that enter the connected Wick pairing arise, and to identify the regime in which the 2D average of Eq. (??) collapses to a 1D convolution (the point-source limit $\Phi = 1$) versus the regime in which the cross-direction carrier-phase factor $\mathcal{F}_{\text{cross}}$ of Eq. (??) appears.

1. Setup

The observed field is Eq. (??),

$$E_{\text{obs}}(t) = \int d\boldsymbol{\theta} I^{1/2}(\boldsymbol{\theta}) E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta})), \quad (\text{A1})$$

with three statistical inputs:

1. Each $E_{\boldsymbol{\theta}}$ is a stationary, zero-mean, complex-Gaussian (analytic-signal) process with unit mean intensity $\langle |E_{\boldsymbol{\theta}}(s)|^2 \rangle = 1$.
2. Source spatial incoherence: $\langle E_{\boldsymbol{\theta}}^*(s) E_{\boldsymbol{\theta}'}(s') \rangle = g_{\text{src}}^{(1)}(s' - s) \delta(\boldsymbol{\theta} - \boldsymbol{\theta}')$, with anomalous pairings $\langle EE \rangle$ and $\langle E^* E^* \rangle$ vanishing.
3. δt is independent of E , and its joint distribution at distinct $(t, \boldsymbol{\theta})$ is the Gaussian process whose covariance is Eq. (4) divided by c^2 .

The brightness distribution is normalized, $\int I(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1$, so $\langle I_{\text{obs}} \rangle = 1$ throughout this appendix.

2. Mean intensity

Plugging Eq. (A1) into $\langle |E_{\text{obs}}(t)|^2 \rangle$ and using spatial incoherence to collapse the angular integral to the diagonal,

$$\begin{aligned} \langle |E_{\text{obs}}(t)|^2 \rangle &= \iint d\boldsymbol{\theta} d\boldsymbol{\theta}' I^{1/2}(\boldsymbol{\theta}) I^{1/2}(\boldsymbol{\theta}') \langle E_{\boldsymbol{\theta}}^*(t - \delta t_1) E_{\boldsymbol{\theta}'}(t - \delta t_2) \rangle \\ &= \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) \langle |E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta}))|^2 \rangle = \int d\boldsymbol{\theta} I(\boldsymbol{\theta}) = 1, \end{aligned} \quad (\text{A2})$$

where $\delta t_1 = \delta t(t; \boldsymbol{\theta})$ and $\delta t_2 = \delta t(t; \boldsymbol{\theta}')$. By stationarity, the random retarded time does not change the marginal second moment of each $E_{\boldsymbol{\theta}}$, so the foam preserves the mean intensity exactly: $\langle |E_{\text{obs}}|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$.

Appendix B: FIRAS Constraint on the Foam Exponent

1. Physical Motivation

The Wiener–Khinchin decomposition of Eq. (26) applies to any stationary field attenuated by the foam scattering kernel $\mathcal{K}_{\text{foam}}(\nu)$, not only to narrow emission lines. The Cosmic Microwave Background (CMB), observed across $\nu \in [6, 600]$ GHz by COBE/FIRAS, provides an independent broadband laboratory for this test: the same two-component signature — a suppressed coherent core and an incoherent pedestal (Section IV) — should appear in the CMB monopole spectrum if spacetime foam distorts photons on their path from the last-scattering surface. Because the foam signal is achromatic (geometric path-length jitter, not plasma dispersion), the frequency dependence of the foam-induced distortion follows $\sigma_{\phi}^2 \propto \nu^2$, distinct from standard chemical-potential (μ) and Compton- y spectral distortions. FIRAS constrains such distortions at the level $|\mu|, |y| \lesssim 10^{-5}$ to 10^{-4} , providing a precision test of foam models in the parameter range that is inaccessible to optical imaging tests.

2. Foam Phase Variance at CMB Frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_C \approx 14$ Gpc (Planck 2018 flat Λ CDM cosmology; $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). Equation (23) gives the accumulated phase variance directly at the comoving distance $D_C \approx 14$ Gpc:

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c}\right)^2 A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (\text{B1})$$

where ν is the observed frequency. The strong ν^2 dependence means foam effects are largest at the high-frequency end of the FIRAS band. Representative values at $\nu = 150$ GHz ($\lambda_{\text{CMB}} \approx 2$ mm):

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 2 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (\text{B2})$$

FIRAS is thus sensitive to foam only in the range $\alpha \lesssim 0.55$, below the holographic value but above the random-walk case (which FIRAS rules out decisively).

3. Fast-Foam Spectral Template

In the dynamical limit $\tau_{\text{foam}} \ll \tau_c(\text{CMB})$, where the CMB photon coherence time at frequency ν is $\tau_c = 1/\nu \simeq 7$ ps at 150 GHz, the foam acts as a fully dynamic phase screen. By the same argument as Section IV.B, the observed CMB specific intensity decomposes into a coherent core and an incoherent pedestal:

$$I_\nu^{\text{obs}}(\alpha) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + K_\nu^{\text{fast}} \int_{\nu_{\min}}^{\nu_{\max}} [1 - e^{-\sigma_\phi^2(\nu')}] B_{\nu'}(T_0) d\nu', \quad (\text{B3})$$

where $B_\nu(T_0)$ is the Planck function at the CMB temperature T_0 , and $K_\nu^{\text{fast}} = [\nu \ln(\nu_{\max}/\nu_{\min})]^{-1}$ is the photon-conserving redistribution kernel for the fast-foam limit, normalized over $[\nu_{\min}, \nu_{\max}] = [6, 600]$ GHz (the FIRAS band). The first term is the coherent core — the fraction $e^{-\sigma_\phi^2}$ of photons that traverse the foam without scattering — and the second term is the incoherent pedestal, distributing the scattered fraction uniformly in $\ln \nu$ over the band. The foam-induced spectral distortion is

$$\Delta I_\nu^{\text{foam}}(\alpha) \equiv I_\nu^{\text{obs}}(\alpha) - B_\nu(T_0), \quad (\text{B4})$$

and photon conservation (constant total flux) is satisfied by construction:

$$\int_{\nu_{\min}}^{\nu_{\max}} \Delta I_\nu^{\text{foam}} d\nu = 0. \quad (\text{B5})$$

For $\sigma_\phi^2 \ll 1$ the distortion is $\Delta I_\nu^{\text{foam}} \approx \sigma_\phi^2(\nu)[-B_\nu(T_0) + K_\nu^{\text{fast}} \langle B \rangle]$, where $\langle B \rangle = \int B_{\nu'} d\nu'$: a high-frequency deficit (since $\sigma_\phi^2 \propto \nu^2$ removes more flux at high ν) compensated by a low-frequency pedestal.

4. Validity of the Fast-Foam Assumption

The fast-foam limit requires $\tau_{\text{foam}} \ll 1/\nu_{\text{CMB}} \sim 7$ ps. If the foam coherence time is set by the Planck scale, $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44}$ s, this condition is satisfied by forty orders of magnitude. For larger τ_{foam} , the foam suppression factor $e^{-\sigma_\phi^2[1-C_\alpha(\tau_{\text{foam}})]}$ (Eq. (24)) is reduced by the factor $[1 - C_\alpha(\tau_{\text{foam}})]$, and the distortion amplitude is suppressed accordingly. In the quasi-static limit $\tau_{\text{foam}} \rightarrow \infty$, $C_\alpha \rightarrow 1$ and the CMB spectrum is unchanged, consistent with the vanishing of the HBT Siegert signal in that limit (Section III).

5. Bayesian Inference

The FIRAS monopole spectrum is compared to the foam-distorted blackbody of Eq. (B4) together with standard spectral-distortion templates (μ -distortion, Compton- y , CIB, and calibration modes) via a Gaussian log-likelihood

$$\ln \mathcal{L} = -\frac{1}{2} \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r}, \quad (\text{B6})$$

where $\mathbf{r} = \mathbf{I}^{\text{obs}} - \mathbf{I}^{\text{model}}$ is the 43-element residual vector and $\mathbf{C}$ is the FIRAS noise-plus-calibration covariance matrix [14]. The free parameters are $(\alpha, A_\alpha, T_0, \mu, y, A_{\text{CIB}}, c_1, c_2, c_3)$, where c_i are calibration mode amplitudes. Sampling uses the No-U-Turn Sampler [15] as implemented in NumPyro, with four chains of 2000 warmup and 2000 sampling steps; convergence is assessed via $\hat{R} < 1.01$ and ESS > 1000 on all parameters. A null-hypothesis fit with $A_\alpha = 0$ must reproduce the published values $T_0 = 2.72548 \pm 0.00057$ K, $|\mu| < 9 \times 10^{-5}$ and $|y| < 1.5 \times 10^{-5}$ (95% CL) [14, 16] before the foam parameter is freed.

6. Connection to the Spectral Framework of Section IV

Equation (B3) is the broadband analogue of Eq. (26): the coherent core $e^{-\sigma_\phi^2} B_\nu(T_0)$ corresponds to the $\delta(\nu)$ term carrying fraction $e^{-\sigma_\phi^2}$ of the source flux, while the incoherent pedestal corresponds to the residual Fourier transform $\mathcal{F}\{\mathcal{F}_{\text{foam}}(\tau) - e^{-\sigma_\phi^2}\}(\nu)$ convolved with the source spectrum. The two-component structure predicted in Section IV.B — with the distinctive high- ν deficit and low- ν pedestal of the fast-foam limit — is what FIRAS constrains, using the full dynamic range of the 6–600 GHz band rather than the narrow velocity window of a single spectral line.

-
- [1] J. A. Wheeler, *Physical Review* **97**, 511 (1955).
 - [2] S. W. Hawking, *Nuclear Physics B* **144**, 349 (1978).
 - [3] S. Carlip, *Rep. Prog. Phys.* **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
 - [4] Y. J. Ng and E. S. Perlman, *Universe* **8**, 382 (2022).
 - [5] Y. J. Ng, *Physical Review Letters* **86**, 2946 (2001).
 - [6] Y. J. Ng, *Modern Physics Letters A* **18**, 1073 (2003).
 - [7] R. Lieu and L. W. Hillman, *The Astrophysical Journal* **585**, L77 (2003).
 - [8] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, *Physical Review D* **83**, 084003 (2011).
 - [9] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, *The Astrophysical Journal* **805**, 10 (2015).
 - [10] V. S. H. Lee, K. M. Zurek, and Y. Chen, *Physical Review D* **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
 - [11] C. Feruglio, E. Daddi, F. Fiore, D. M. Alexander, E. Piconcelli, and C. Malacaria, *Astrophysical Journal Letters* **729**, L4 (2011), arXiv:1101.3478.
 - [12] G. P. Szokoly, J. Bergeron, G. Hasinger, I. Lehmann, L. Kewley, V. Mainieri, M. Nonino, P. Rosati, R. Giacconi, R. Gilli, R. Gilmozzi, C. Norman, M. Romaniello, E. Schreier, P. Tozzi, J. X. Wang, W. Zheng, and A. Zirm, *Astrophysical Journal Supplement Series* **155**, 271 (2004), arXiv:astro-ph/0312141.
 - [13] E. Vanzella, S. Cristiani, M. Dickinson, M. Giavalisco, H. Kuntschner, J. Haase, M. Nonino, P. Rosati, C. Cesarsky, H. C. Ferguson, R. A. E. Fosbury, A. Grazian, L. A. Moustakas, A. Rettura, P. Popesso, A. Renzini, D. Stern, and GOODS Team, *Astronomy & Astrophysics* **454**, 423 (2006), arXiv:astro-ph/0509665.
 - [14] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright, *Astrophysical Journal* **473**, 576 (1996), arXiv:astro-ph/9605054.
 - [15] M. D. Hoffman and A. Gelman, *Journal of Machine Learning Research* **15**, 1593 (2014), arXiv:1111.4246.
 - [16] D. J. Fixsen, *Astrophysical Journal* **707**, 916 (2009), arXiv:0911.1955.